

The Reverse Regression Fallacy

Research Design Module 1

Beaver, Lambert, and Ryan (1987), *Journal of Accounting and Economics*

ACCTG 597A · Fall 2026 · Sam Bonsall · Penn State

One sample. One parameter. Two answers.

BLR (1987) estimate the same structural parameter two ways, on the same data:

Forward regression of returns on earnings

0.31

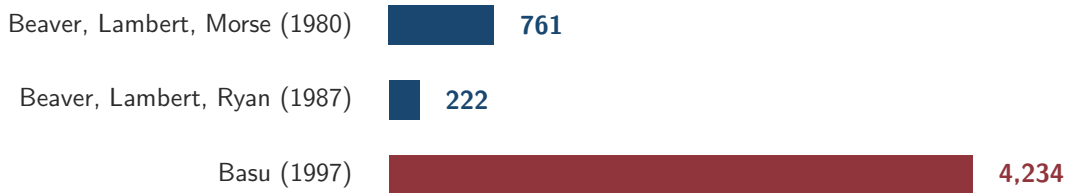
Implied by their reverse-regression algebra

2.38

A factor of 7.7 apart.

Which one is wrong, and why did the literature choose it?

The flawed direction became infrastructure



Citations (OpenAlex, August 2026)

The earnings-on-returns direction anchors conditional conservatism, one of the largest empirical literatures in accounting.

We take up Basu (1997) in Week 4. Today is about the econometrics it inherited.

BLR's claim: reversing the regression fixes errors-in-variables

$$\underbrace{G_i = \gamma g_i + u_i}_{\text{forward: } g \text{ correlated with } u} \implies \underbrace{g_i = \delta G_i + e_i}_{\text{"reverse": error moved to } e}$$

"The reverse regression approach places the measurement error in the disturbance term, rather than in the explanatory variable... As a result, the estimate of the slope coefficient is not expected to be biased."

BLR (1987, p. 143)

The promised free lunch: no grouping, no bias, no efficiency loss.

BLR's own Property (3) contains the refutation

Section 2.2 lists three “well documented” properties (citing Malinvaud 1970):

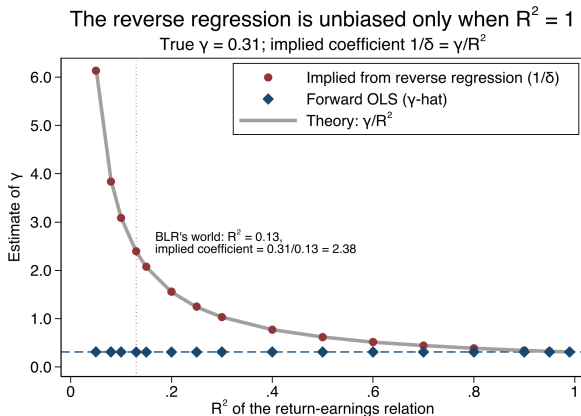
$$\gamma = \frac{\text{cov}(g, G)}{\text{var}(g)}, \quad \delta = \frac{\text{cov}(g, G)}{\text{var}(G)}, \quad \gamma\delta = R^2 \implies \frac{1}{\delta} = \frac{\gamma}{R^2}$$

The reverse regression recovers the forward coefficient **only when $R^2 = 1$** .

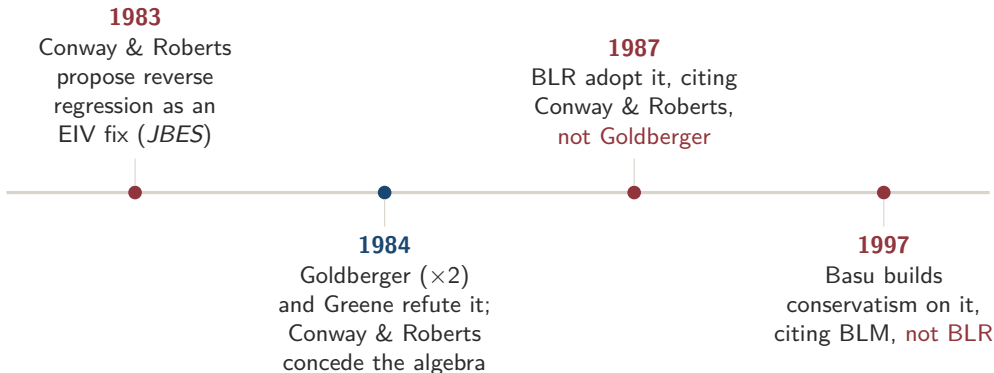
At the return–earnings relation's actual $R^2 \approx 0.13$, the “fix” inflates the coefficient by $1/0.13 \approx 7.7\times$.

They derived the inflation factor themselves, then proceeded anyway.

At $R^2 = 0.13$, the “corrected” coefficient is inflated $7.7\times$



The econometrics said no before the accounting said yes



The refutation ran in the *same journal* as the paper BLR cite. Only one accounting paper has ever cited both BLR and Goldberger: Dietrich, Muller, and Riedl (2007).

Observed earnings are not measured with error

What classical EIV requires

An *observed* regressor that mismeasures a well-defined true quantity, with error uncorrelated with everything else.

What BLR actually have

Earnings measured *exactly*. The “garbling” afflicts a latent construct — economic income — defined only inside their model.

Meanwhile the dependent variable is contaminated: returns carry discount-rate news, growth options, and sentiment, none of it current earnings.

“To the extent that $\Delta P_{it}/P_{i,t-1}$ changes due to other factors . . . such variation can be viewed as a form of measurement error.”

BLR (1987, p. 151), conceding the condition their claim requires

Two economies, one $\hat{\delta}$: the reverse coefficient is not identified

BLR read $\hat{\delta}$ as $1/(1 - \theta)$, where θ is the transitory share of earnings variance. Simulate two economies calibrated to their moments:

	World A transitory earnings, clean returns	World B no transitory earnings, noisy returns
True transitory share (θ)	69%	0%
Forward $\hat{\gamma}$	0.309	0.308
R^2	0.130	0.128
Reverse $\hat{\delta}$	0.420	0.416

$\hat{\delta}$ cannot distinguish transitory earnings from non-earnings return variation.
It is a statement about variance shares, not a structural parameter.

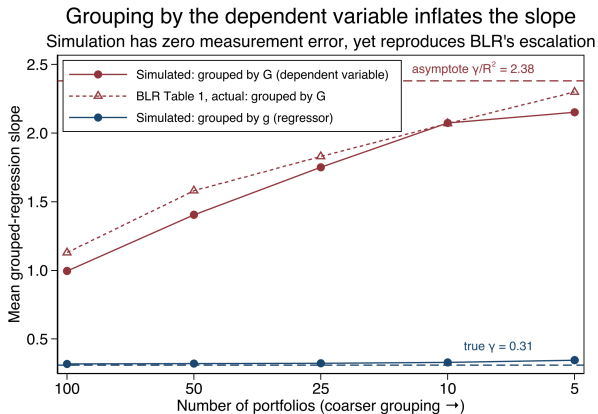
Grouping by the dependent variable sorts on the error term

Wald (1940), Madansky (1959): grouping is consistent only if assignment is *independent of the errors*. Portfolios with high G are portfolios whose u came in high (Pakes 1982).

```
set obs 700
gen g = rnormal(0,1)
gen G = 0.31*g + rnormal(0,sqrt(0.643)) // NO measurement error
xtile port = G, nq(5) // group by the DEPENDENT variable
collapse (median) G g, by(port)
reg G g // slope: about 2.2, not 0.31
```

Full simulation: github.com/sbb151/acctg597a-research-design (base Stata, runs in ~1 minute)

A world with zero measurement error reproduces BLR's Table 1



BLR's own specification tests rejected the framework

- Grouped-by- g coefficient reaches **2.14**; with their $\hat{\delta} = 0.59$, the theory predicts $0.59/0.13 = \mathbf{4.54}$
- Grouped-by- G reverse coefficients **decline** in all 19 sample years; the theory says flat
- Alternative price variables (raw and residual returns) make the anomalies *worse*

"None of the above tests is particularly successful in resolving the possible misspecification . . . However, the next step is not immediately obvious."

BLR (1987, p. 151)

The bounds defense holds only in the textbook EIV model

“Forward and reverse regressions bound the true coefficient.”

Only in the classical one-regressor EIV model (Klepper and Leamer 1984). Returns carry their own large non-earnings noise, so the bounding model does not apply to the return–earnings setting.

“BLR derived the grouping algebra correctly, so the paper polices itself.”

Deriving that $\hat{\gamma}_p \rightarrow \gamma/R^2$ and interpreting $1/\hat{\delta}$ as a structural parameter are different acts; BLR did the first and built the paper on the second, and their own specification tests rejected the framework.

Summary

- The reverse regression estimates a different parameter from the forward regression; BLR's own Property (3) implies the two coincide only when R^2 equals one.
- Grouping by the dependent variable induces correlation between the regressor and the disturbance, so the coefficient inflation is mechanical and is not evidence of measurement error.
- The econometrics literature documented both results before the design entered accounting (Goldberger 1984); evaluating a research design requires reading beyond the literature that adopted it.

We return to the design's most successful descendant, Basu (1997), in Week 4.